

```
int main(void)
{
    Date myDate(5, 10, 2025);
}
```

Date myDate; // This statement will result into 12 bytes allocation.

Rule: If T is a data type then there are two ways to create an instance of type T.

1) Use data definition statement:

```
T variable_name;
T variable_name(initialization_data);
```

2) Use dynamic memory allocation function:

C:

```
T* pT = (T*)malloc(sizeof(T));
```

C++:

```
T* pT = new T;
```

```
T myVar; // an instance of type T is allocated and its named as myVar
        // &myVar
```

```
int num;
int num = 10;
```

```
struct Date myDate;
struct Date myDate = {5, 10, 2025};
```

```
Date myDate(5, 10, 2025);
```

1) &myDate : Address of object to be initialized
2) 5 : initial value of day
3) 10 : initial value of month
4) 2025 : initial value of year

// At server side: at class definition side

```
class Date
{
    public:
        Date(Date* pDate, int init_day, int init_month, int init_year)
        {
            pDate->day = init_day;
```

```
        pDate->month = init_month;
        pDate->year = init_year;
    }
};
```

```
Student firstStudent("Yogeshwar", 134, 98.9, 99.9);
```

- 1) &firstStudent: Address of object of class Student
- 2) "Yogeshwar": Name of student
- 3) 134: Roll number of student
- 4) 98.9: Marks of student
- 5) 99.9: Attendance of student

```
class Student
{
    private:
        char* st_name;
        int st_roll;
        float st_marks;
        float st_attendance;
    public:
        Student(Student* pStudent, char* name, int roll_number, float marks,
                float attendance)
        {
            pStudent->st_name = name;
            pStudent->st_roll = roll_number;
            pStudent->st_marks = marks;
            pStudent->st_attendance = attendance;
        }
};
```

```
class T
{
    public:
        T()
        {

        }
};
```

```
class Date
{
    private:
        int day, month, year;
    public:
        Date()
        {
            this->day = 5;
            this->month = 10;
            this->year = 2025;
        }
};
```

```
class Date
{
    private:
        int day, month, year;
    public:
        Date(Date* p)
        {
            p->day = 5;
            month = 10;
            year = 2025;
        }
};
```